

Power Management ICs Enable More Features At Less Power

Packing more and more functionalities yet keeping the power consumption low has been the *mantra* followed by electronic product designers. This has led to a lot of new trends in designing a power management system that keeps power consumption in control



Freescale MC13783—a highly integrated power management, audio and user interface component

■ SHWETA DHADIWAL BAID

Back in 1990s when mobile phones where commercially launched in India, they required charging every day to serve the basic functionality of calling. Today's cell-phones have ten times more functionalities but require charging only once in four days. This is what product designers are trying to achieve—more functions requiring lesser power.

Power management is a universal concept. Every designer needs to understand why and how to manage power. "If the power is not managed properly, it will degrade the system performance," shares Jimmy Chalakkal, FAE manager, Analog Devices. "You may have the best possible designers developing a very high-performance

system, but if power management is not taken into consideration, it will generate a lot of other losses, degrading the overall system performance," he adds.

"Power management means power conversion at elevated efficiencies," explains Ram Anant, director for power management products, Texas Instruments. "As the efficiency increases, power consumption reduces." Using less power but delivering high performance is the design trend followed by the designer community. To achieve this, a designer has to trade-off between four main parameters—cost, efficiency, form factor and component count.

Power budgeting

Power budget is allocation of available power among various functions of the system. You need to analyse the power

requirement using simulation and assign appropriate power level to every sub-system or function. As a designer, you need to consider the performance of the overall system rather than individual component.

"Earlier in the industry, designers used to select the component first, then check individual performance of the component, and build a system that met or exceeded system-level performance," says Chalakkal. Engineers often 'over spec' their system power budget by selecting the most conservative estimations so that the error margin is less. However, by doing so, the overall cost of product goes high due to increased packaging and cooling requirements.

Today, we see a reversal in trend. Says Chalakkal, "Designers are following a much more systematic approach today. They do power budgeting in advance and choose high-performance converters and amplifiers that meet the power budget." Better power estimation gives the same result at low cost with much more efficient design.

Integrated power management semiconductors

"Just as your water tap leaks after closing, all integrated circuits (ICs) leak current even if they are not functioning," explains Anant. If you can control leakage current in all the ICs, every micro- and milli-ampere adds up to a large amount. The system architecture should be designed such that if a certain part of the system is not in use, it can be turned off completely.